(v) **Binomial Surd :** A binomial surd is an algebraic sum (or difference) of 2 terms both of which could be surds or one could be a rational number and another a surd. For example, $\frac{1}{2}-\sqrt{19}, 5+3\sqrt{2}, \sqrt{3}-2\sqrt{7}$ are binomial surds.

Example 2.20 Arrange in ascending order: $\sqrt[3]{2},\ \sqrt[2]{4},\ \sqrt[4]{3}$

Solution

The order of the surds $\sqrt[3]{2},\ \sqrt[2]{4}$ and $\sqrt[4]{3}$ are 3, 2, 4.

L.C.M. of 3, 2, 4 = 12.

$$\sqrt[3]{2}=\left(2^{\frac{1}{3}}\right)=\left(2^{\frac{4}{12}}\right)=\sqrt[12]{2^4}=\sqrt[12]{16}\ ;\qquad \sqrt[2]{4}=\left(4^{\frac{1}{2}}\right)=\left(4^{\frac{6}{12}}\right)=\sqrt[12]{4^6}=\sqrt[12]{4096}$$

$$\sqrt[4]{3}=\left(3^{\frac{1}{4}}\right)=\left(3^{\frac{3}{12}}\right)=\sqrt[12]{3^3}=\sqrt[12]{27}$$

The ascending order of the surds $\sqrt[3]{2},\ \sqrt[4]{3},\ \sqrt[2]{4}$ is $\sqrt[12]{16}<\sqrt[12]{27}<\sqrt[12]{4096}$

that is, $\sqrt[3]{2}\ ,\ \sqrt[4]{3}\ ,\ \sqrt[2]{4}\ .$

2.6.2 Laws of Radicals

For positive integers m, n and positive rational numbers a and b, it is worth remembering the following properties of radicals:

S.No.	Radical Notation	Index Notation
1.	$\left(\sqrt[n]{a}\right)^n=a=\sqrt[n]{a^n}$	$\left(a^{\frac{1}{n}}\right)^n=a=\left(a^n\right)^{\frac{1}{n}}$
2.	$\sqrt[n]{a}\times\sqrt[n]{b}=\sqrt[n]{ab}$	$a^{\frac{1}{n}}\times b^{\frac{1}{n}}=\left(ab\right)^{\frac{1}{n}}$
3.	$\sqrt[m]{\sqrt[n]{a}}=\sqrt[mn]{a}=\sqrt[n]{\sqrt[m]{a}}$	$\left(a^{\frac{1}{n}}\right)^{\frac{1}{m}}=a^{\frac{1}{mn}}=\left(a^{\frac{1}{m}}\right)^{\frac{1}{n}}$
4.	$\frac{\sqrt[n]{a}}{\sqrt[n]{b}}=\sqrt[n]{\frac{a}{b}}$	$\frac{a^{\frac{1}{n}}}{b^{\frac{1}{n}}}=\left(\frac{a}{b}\right)^{\frac{1}{n}}$

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We shall now discuss certain problems which require the laws of radicals for simplifying.

Example 2.21 Express each of the following surds in its simplest form (i) $\sqrt[3]{108}$ (ii) $\sqrt[3]{\left(1024\right)^{-2}}$ and find its order, radicand and coefficient.

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